

BRST ANALYSIS OF QCD_2 AS A PERTURBED WZW THEORY

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Received 31 July 1995

Revised 30 October 1995

The integrability of quantum chromodynamics in $1 + 1$ dimensions has recently been suggested by formulating it as a perturbed conformal Wess–Zumino–Witten theory. This paper further elucidates this formulation by presenting a detailed BRST analysis.

1. Introduction

As the non-Abelian extension of the exactly soluble Schwinger model,¹ quantum chromodynamics in $1 + 1$ dimensions (QCD_2) has received much attention in the past 25 years. Its equivalent bosonic description was, however, fully understood only as late as 1984, with the work of Polyakov–Wiegmann² and Witten.³

The basic idea for arriving at an equivalent bosonic description of QCD_2 is to perform a chiral transformation of the fermion fields in the partition function, which decouples the fermions from the gauge field (decoupled picture). This idea, explored in Ref. 4 for the bosonization of the Schwinger model,¹ has been successfully applied in a number of papers on QCD_2 .^{5,6} For a general review the reader is referred to Ref. 7. Following this procedure one arrives at an effective action involving a Wess–Zumino–Witten (WZW) functional of the gauge field “potentials” plus the Yang–Mills action.

Choosing the gauge field “potentials” as integration variables, E. Abdalla and M. C. B. Abdalla arrived at a representation of the QCD_2 partition function as the

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product of a conformally invariant sector involving positive and negative level WZW fields as well as ghosts, and a massive, conformally noninvariant sector. Although the various sectors appear to be decoupled on the Lagrangian level, they were shown to be connected by first and second class constraints. These constraints are in fact necessary in order to avoid negative metric states, associated with the negative level WZW fields, in the physical Hilbert space.

The first class constraints we expect to be associated with (nilpotent) BRST charges. The physical states are required to be annihilated by these charges. For a complete characterization of the physical Hilbert space of QCD₂ in the formulation of Ref. 8 we thus need to construct the BRST currents from the corresponding BRST symmetries. This is the objective of this paper. We perform the construction for both the “local” and “nonlocal” formulation of QCD₂, and obtain in each case two conserved BRST currents corresponding to left- and right-moving currents, just as in a conformally invariant WZW theory. The standard form of these currents sheds further light on the role of the first class constraints obtained in Ref. 8.

It turns out that the BRST charges in the nonlocal formulation provide a particularly transparent view of the restrictions to be imposed on the physical states, since they are found to manifestly operate only on the conformally invariant sector of the theory. The implications of this for the degeneracy of the QCD₂ ground state are the subject matter of Ref. 16.

2. Local, Decoupled Formulation of QCD₂

The partition function of QCD₂ is given by

$$Z = \int [\mathcal{D}A_\mu] \int \mathcal{D}\psi \mathcal{D}\bar{\psi} \exp[i(S_{\text{YM}} + S_F)], \quad (2.1)$$

where S_{YM} and S_F are the Yang–Mills and fermionic action, respectively:^a

$$S_{\text{YM}} = -\frac{1}{4} \int d^2x \operatorname{tr}(F_{\mu\nu} F^{\mu\nu}), \quad (2.2)$$

$$\begin{aligned} S_F &= \int d^2x \bar{\psi}(i\partial + eA)\psi \\ &= \int d^2x \{ \psi_1^\dagger (i\partial_+ + eA_+) \psi_1 + \psi_2^\dagger (i\partial_- + eA_-) \psi_2 \} \end{aligned} \quad (2.3)$$

and $[\mathcal{D}A_\mu]$ stands for the measure including gauge fixing. It will be convenient to work in the light cone gauge $A_+ = (A_0 + A_1) = 0$. We implement this gauge in terms of a Lautrup–Nakanishi–Lagrange multiplier field B ,

^aOur conventions are $\partial_\pm = \partial_0 \pm \partial_1$, $A_\pm = A_0 \pm A_1$, $F_{\mu\nu} = t^a F_{\mu\nu}^a$, etc. with the normalization $\operatorname{tr} t^a t^b = \delta^{ab}$ for the Hermitian generators in the fundamental representation, and the commutation relation $[t^a, t^b] = ifabc t^c$, with $fabcfabd = \frac{C_V}{2} \delta cd$. We follow in general the notation and conventions of Ref. 8.

$$Z = \int \mathcal{D}A_\mu \int \mathcal{D}\psi \mathcal{D}\bar{\psi} \int \mathcal{D}B \int \mathcal{D}b_- \mathcal{D}c_- \exp(iS_{\text{GF}}) \quad (2.4)$$

with

$$S_{\text{GF}} = S_{\text{YM}} + S_F + S_{\text{gh}} + \int d^2x \operatorname{tr}(BA_+), \quad (2.5)$$

where S_{gh} is the ghost action

$$S_{\text{gh}} = \int d^2x \operatorname{tr}(b_- i\mathcal{D}_+ c_-) \quad (2.6)$$

with

$$\mathcal{D}_\pm = \partial_\pm - ie[A_\pm,] \quad (2.7)$$

the covariant derivative in the adjoint representation.

The gauge-fixed action (2.5) is invariant under the BRST transformation

$$\begin{aligned} \delta A_\mu &= \epsilon \frac{1}{e} \mathcal{D}_\mu c_-, \\ \delta \psi_\alpha &= \epsilon c_- \psi_\alpha, \\ \delta c_- &= \epsilon \frac{1}{2} \{c_-, c_-\}, \\ \delta b_- &= \epsilon \frac{1}{e} B, \\ \delta B &= 0, \end{aligned} \quad (2.8)$$

with ϵ a Grassman-valued infinitesimal parameter.

Performing the integration over B and A_+ , we obtain the partition function

$$Z = \int \mathcal{D}A_- \int \mathcal{D}\psi_1^{(0)} \mathcal{D}\psi_1^{\dagger(0)} \int \mathcal{D}\psi_2 \mathcal{D}\psi_2^\dagger \int \mathcal{D}b_-^{(0)} \mathcal{D}c_-^{(0)} \exp(iS_{\text{GF}}) \quad (2.9)$$

with the corresponding gauge-fixed Lagrangian

$$\mathcal{L}_{\text{GF}} = \operatorname{tr} \frac{1}{8} (\partial_+ A_-)^2 + \psi_1^{(0)\dagger} i \partial_+ \psi_1^{(0)} + \psi_2^\dagger i D_- \psi_2 + \operatorname{tr}(b_-^{(0)} i \partial_+ c_-^{(0)}). \quad (2.10)$$

We have denoted the fields obeying a free field dynamics by a superscript “(0),” and D_- is the covariant derivative

$$D_\pm = \partial_\pm - ieA_\pm \quad (2.11)$$

with $A_\pm$ in the fundamental representation.

The classical action S_{GF} is invariant under the BRST transformation

$$\begin{aligned} \delta A_- &= -i\epsilon \frac{1}{e} \mathcal{D}_- c_-^{(0)}, \\ \delta \psi_1^{(0)} &= \epsilon c_-^{(0)} \psi_1^{(0)}, \\ \delta \psi_2 &= \epsilon c_-^{(0)} \psi_2, \\ \delta c_-^{(0)} &= \epsilon \frac{1}{2} \{c_-^{(0)}, c_-^{(0)}\}, \end{aligned} \quad (2.12)$$

$$\delta b_-^{(0)} = \epsilon \frac{1}{e^2} \mathcal{D}_-(\partial_+ A_-) + \epsilon \psi_1^{(0)} \psi_1^{(0)\dagger} + \epsilon \{b_-^{(0)}, c_-^{(0)}\}. \quad (2.13)$$

These transformation laws can be obtained from (2.8) by eliminating A_+ and B using the equations of motion

$$\begin{aligned} A_+ &= 0, \\ B^a &= -\frac{\delta S_{\text{YM}}}{\delta A_+^a} - e \psi_1^\dagger t^a \psi_1 + i e f_{abc} b_-^b c_-^c. \end{aligned} \quad (2.14)$$

Regarding (2.12) as a classical transformation, it is a symmetry of the gauge-fixed Lagrangian (2.10). In order to analyze the corresponding symmetry at the partition function level, one has however to take into account the following facts:

- (i) The fermionic path integral in (2.4) (i.e. before fixing the gauge to $A_+ = 0$) leads to

$$\begin{aligned} Z_{\text{fer}} &= \det(i\partial_+ + eA_+) \det(i\partial_- + eA_-) \\ &\times \exp\left(-i \frac{e^2}{4\pi} \int d^2x \operatorname{tr} A_+ A_- \right). \end{aligned} \quad (2.15)$$

The last term in the r.h.s. of (2.15) has been added, exploiting regularization ambiguities, so as to ensure gauge invariance.²

- (ii) Analogously, the determinant arising from the integration of ghosts with action (2.6) has to be adjusted with the same type of $A_+ A_-$ counterterm,

$$Z_{\text{ghosts}} = \det^{Adj}(i\partial_+ + eA_+) \times \exp\left(-i\alpha \int d^2x \operatorname{tr} A_+ A_- \right). \quad (2.16)$$

Since it is well known,² that the determinant in the adjoint representation is related to that in the fundamental through the Casimir C_V (see Ref. 6 for details), consistency of the regularization implies that α should be chosen as $\alpha = (e^2/4\pi)C_V$.

Taking into account (i) and (ii), the transformation law for the ghost field $b_-^{(0)}$ takes the form

$$\delta b_-^{(0)} = -\frac{\epsilon}{e} \mathcal{D}_-(\partial_+ A_-) + \epsilon \psi_1^{(0)} \psi_1^{(0)\dagger} + \epsilon \{b_-^{(0)}, c_-^{(0)}\} - \epsilon e \frac{(1 + C_V)}{4\pi} A_- . \quad (2.17)$$

The last term in the r.h.s. of (2.17) arises from the $A_+ A_-$ counterterms as discussed above. The transformations (2.12) and (2.17) now represent a symmetry transformation of the partition function associated with (2.10).

We now go to a new set of variables such that the partition function factorizes in terms of decoupled fields, by writing

$$A_- = \frac{i}{e} V \partial_- V^{-1}, \quad \psi_2 = V \psi_2^{(0)}. \quad (2.18)$$

For the corresponding transformation of the integration measure one has

$$\begin{aligned}\mathcal{D}A_- &= \int \mathcal{D}b_+ \mathcal{D}c_+ \exp \left(i \int d^2x \operatorname{tr} b_+ i \mathcal{D}_- c_+ \right) \times \mathcal{D}V \\ &= \exp(-iC_V \Gamma[V]) \int \mathcal{D}b_+^{(0)} \mathcal{D}c_+^{(0)} \exp \left(i \int d^2x \operatorname{tr} b_+^{(0)} i \mathcal{D}_- c_+^{(0)} \right) \times \mathcal{D}V, \end{aligned} \quad (2.19)$$

$$\mathcal{D}\psi_2 \mathcal{D}\psi_2^\dagger = \mathcal{D}\psi_2^{(0)} \mathcal{D}\psi_2^{(0)\dagger} \exp(-i\Gamma[V]), \quad (2.20)$$

where $\Gamma[g]$ is the Wess–Zumino–Witten functional³

$$\Gamma[g] = \frac{1}{8\pi} \int d^2x \operatorname{tr} \partial_\mu g^{-1} \partial^\mu g + \frac{1}{12\pi} \int d^3y \epsilon^{\alpha\beta\gamma} \operatorname{tr} [g^{-1} \partial_\alpha g g^{-1} \partial_\beta g g^{-1} \partial_\gamma g] \quad (2.21)$$

with the remarkable property (see Ref. 7 for details)

$$\delta\Gamma[g] = \frac{1}{4\pi} \int d^2x \operatorname{tr} g \delta g^{-1} \partial_+ (g \partial_- g^{-1}) = \frac{1}{4\pi} \int d^2x \operatorname{tr} g^{-1} \delta g \partial_- (g^{-1} \partial_+ g). \quad (2.22)$$

In terms of the new variables, the partition function reads

$$Z = Z_F^{(0)} Z_{\text{gh}}^{(0)} Z_V, \quad (2.23)$$

where

$$Z_V = \int \mathcal{D}V \exp \left\{ -i(1 + C_V) \Gamma[V] + \frac{i}{8e^2} \int d^2x \operatorname{tr} [\partial_+ (V i \mathcal{D}_- V^{-1})]^2 \right\} \quad (2.24)$$

and

$$Z_F^{(0)} = \int \mathcal{D}\psi^{(0)} \mathcal{D}\bar{\psi}^{(0)} \exp \left(i \int d^2x \bar{\psi}^{(0)} i \not{\partial} \psi^{(0)} \right), \quad (2.25)$$

$$Z_{\text{gh}}^{(0)} = \int \mathcal{D}b_\pm^{(0)} \mathcal{D}c_\pm^{(0)} \exp \left[i \int d^2x \operatorname{tr} (b_+^{(0)} i \mathcal{D}_- c_+^{(0)} + b_-^{(0)} i \mathcal{D}_+ c_-^{(0)}) \right]. \quad (2.26)$$

Notice that the WZW action enters in (2.24) with negative level $-(1 + C_V)$.

It is interesting at this stage to compare our results, summarized in Eqs. (2.23)–(2.26), with those presented in Ref. 8. Equation (2.23) shows that the QCD₂ partition function factorizes into the partition functions for free fermions, ghosts and perturbed WZW fields. This factorization (including the remanence of free fermions) is characteristic of path integral bosonization which is always based on the decoupling of the interacting fermions, which thus become free^{4–6} (of course, these free fermions can in turn be bosonized in terms of Wess–Zumino fields $-\tilde{g}$ in Ref. 8).

In terms of the new variables the BRST symmetry transformation (2.12), (2.17) reads

$$\begin{aligned} V\delta V^{-1} &= -\epsilon c_-^{(0)}, \\ \delta\psi_1^{(0)} &= \epsilon c_-^{(0)}\psi_1^{(0)}, & \delta\psi_2^{(0)} &= 0, \\ \delta c_-^{(0)} &= \frac{\epsilon}{2}\{c_-^{(0)}, c_-^{(0)}\}, & \delta c_+^{(0)} &= 0, \\ \delta b_-^{(0)} &= \epsilon B_-^{(0)} + \epsilon\Delta_-(V), & \delta b_+^{(0)} &= 0, \end{aligned} \quad (2.27)$$

where

$$\begin{aligned} B_-^{(0)} &= \psi_1^{(0)}\psi_1^{(0)\dagger} + \{b_-^{(0)}, c_-^{(0)}\}, \\ \Delta_-(V) &= -\frac{1}{4e^2}\mathcal{D}_-(V)(\partial_+(Vi\partial_-V^{-1})) - \left(\frac{1+C_V}{4\pi}\right)Vi\partial_-V^{-1} \end{aligned} \quad (2.28)$$

and

$$\mathcal{D}_-(V) = \partial_- + [V\partial_-V^{-1}, \quad]. \quad (2.29)$$

Using (2.22) one readily checks that the partition function (2.23) is invariant under the above transformation. The corresponding BRST current, as obtained via the usual Noether construction, is found to be [the superscript “(B)” stands for “BRST”]

$$\begin{aligned} J_-^{(B)} &= \text{tr } c_-^{(0)} \left[-\frac{1}{4e^2}\mathcal{D}_-(V)\partial_+(Vi\partial_-V^{-1}) \right. \\ &\quad \left. - \left(\frac{1+C_V}{4\pi}\right)Vi\partial_-V^{-1} + \psi_1^{(0)}\psi_1^{(0)\dagger} + \frac{1}{2}\{b_-^{(0)}, c_-^{(0)}\} \right] \end{aligned} \quad (2.30)$$

with

$$\partial_+ J_-^{(B)} = 0. \quad (2.31)$$

Remarkably enough, BRST symmetry leads to a current which only depends on the variable x^- .

It is desirable to rewrite (2.30) in standard form, exhibiting explicitly its BRST character. To this end we observe that

$$\Omega_- := -\frac{1}{4e^2}\mathcal{D}_-(V)\partial_+(Vi\partial_-V^{-1}) - \left(\frac{1+C_V}{4\pi}\right)Vi\partial_-V^{-1} + j_- \approx 0 \quad (2.32)$$

with

$$j_- = \psi_1^{(0)}\psi_1^{(0)\dagger} + \{b_-^{(0)}, c_-^{(0)}\}, \quad \partial_+ j_- = 0 \quad (2.33)$$

is a constraint of the theory. To see this, we follow the general ideas outlined in Ref. 15, “and gauge” the partition function (2.23) with an external field $W_+ = i\omega^{-1}\partial_+\omega$, by making the substitutions

$$\begin{aligned} \partial_+ &\rightarrow D_+(\omega) = \partial_+ - iW_+, \\ \partial_+ &\rightarrow \mathcal{D}_+(\omega) = \partial_+ - i[W_+, \quad], \end{aligned} \quad (2.34)$$

in the right-hand sector of (2.25) and (2.26), as well as the substitution

$$\Gamma[V] \rightarrow \Gamma[V] - \frac{1}{4\pi} \int \text{tr } W_+ V i \partial_- V^{-1}, \quad (2.35)$$

$$\text{tr}[\partial_+(V \partial_- V^{-1})]^2 \rightarrow \text{tr}[\mathcal{D}_+(\omega) V \partial_- V^{-1} + i \partial_- W_+]^2 \quad (2.36)$$

in (2.24). Noting that

$$\text{tr}[\mathcal{D}_+(\omega) V \partial_- V^{-1} + i \partial_- W_+]^2 = \text{tr}[\partial_+(\omega V) \partial_-(\omega V^{-1})]^2 \quad (2.37)$$

and making use of the Polyakov–Wiegmann identity²

$$\Gamma[gh] = \Gamma[g] + \Gamma[h] + \frac{1}{4\pi} \int d^2x \text{tr}(g^{-1} \partial_+ g h \partial_- h^{-1}), \quad (2.38)$$

one has, after a change of integration variable $V \rightarrow \omega V$,

$$\begin{aligned} Z_F^{(0)} &\rightarrow Z_F^{(0)} e^{-i\Gamma[\omega]}, \\ Z_{\text{gh}}^{(0)} &\rightarrow Z_{\text{gh}}^{(0)} e^{-iC_V \Gamma[\omega]}, \\ Z_V &\rightarrow Z_V e^{i(1+C_V)\Gamma[\omega]}. \end{aligned} \quad (2.39)$$

This shows that the partition function (2.23) is unchanged by the transformations (2.34) to (2.36). From here we derive the constraint (2.32) by taking the functional derivative of the gauged partition function with respect to W_+ , and setting $W_+ = 0$. This constraint (Gauss' law) can be shown to satisfy a Kac–Moody algebra^b with vanishing central charge; hence, in the terminology of Dirac,¹⁰ it is first class.

In terms of the constraint (2.32), the current (2.30) takes the standard form expected from general considerations:^{11,12}

$$J_-^{(B)} = \text{tr} \left[c_-^{(0)} \Omega_- - \frac{1}{2} b_-^{(0)} \{c_-^{(0)}, c_-^{(0)}\} \right]. \quad (2.40)$$

Since Ω_- is first class, the corresponding charge $Q_-^{(B)}$ is nilpotent.

A second BRST symmetry

As is well known,^{13,14} one expects a further BRST current associated with the change of variables (2.18). In fact, it is easy to see that the partition function (2.23) is also invariant under the transformation

$$\begin{aligned} V^{-1} \delta V &= -\epsilon c_+^{(0)}, \\ \delta \psi_1^{(0)} &= 0, \quad \delta \psi_2^{(0)} = \epsilon c_+^{(0)} \psi_2^{(0)}, \\ \delta c_-^{(0)} &= 0, \quad \delta c_+^{(0)} = \frac{\epsilon}{2} \{c_+^{(0)}, c_+^{(0)}\}, \\ \delta b_-^{(0)} &= 0, \quad \delta b_+^{(0)} = \epsilon B_+^{(0)} + \epsilon \Delta_+(V), \end{aligned} \quad (2.41)$$

^bIn Ref. 9 the terminology “affine Lie algebra” is preferred.

with $B_+^{(0)}$ and $\Delta_+(V)$ given by

$$\begin{aligned} B_+^{(0)} &= \psi_2^{(0)} \psi_2^{(0)\dagger} + \{b_+^{(0)}, c_+^{(0)}\}, \\ \Delta_+(V) &= \frac{1}{e^2} V^{-1} (\partial_+^2 (V i \partial_- V^{-1})) V - \frac{1+C_V}{4\pi} V^{-1} i \partial_+ V. \end{aligned} \quad (2.42)$$

This transformation law should be compared with the one in (2.27). The corresponding BRST current obtained via the standard Noether construction is found to be

$$\begin{aligned} J_+^{(B)} &= \text{tr } c_+^{(0)} \left[\frac{1}{4e^2} V^{-1} (\partial_+^2 (V i \partial_- V^{-1})) V \right. \\ &\quad \left. - \frac{1+C_V}{4\pi} V^{-1} i \partial_+ V + \psi_2^{(0)} \psi_2^{(0)\dagger} + \frac{1}{2} \{b_+^{(0)}, c_+^{(0)}\} \right]. \end{aligned} \quad (2.43)$$

To put this expression into standard form, we observe, following again the method of Ref. 15, that gauging with the external field,

$$W_- = \frac{i}{e} \omega \partial_- \omega^{-1} \quad (2.44)$$

by making the substitutions

$$\partial_- \rightarrow \partial_- - i W_-, \quad \partial_- \rightarrow \partial_- - i [W_-, \quad], \quad (2.45)$$

in the left-hand sector of (2.25) and (2.26), we have in analogy to (2.39),

$$\begin{aligned} Z_F^{(0)} &\rightarrow Z_F^{(0)} e^{-i\Gamma[\omega]}, \\ Z_{\text{gh}}^{(0)} &\rightarrow Z_{\text{gh}}^{(0)} e^{-iC_V \Gamma[\omega]}, \\ e^{-i(1+C_V)\Gamma[V]} &\rightarrow e^{-i(1+C_V)(\Gamma[V\omega] - \Gamma[\omega])}, \\ Vi\partial_- V^{-1} &\rightarrow V(i\partial_- + W_-)V^{-1} = (V\omega)i\partial_- (V\omega)^{-1}. \end{aligned} \quad (2.46)$$

Hence the partition function (2.23) is left invariant by this transformation. From this, we can derive the constraint

$$\Omega_+ \equiv \frac{1}{4e^2} V^{-1} [\partial_+^2 (V i \partial_- V^{-1})] V - \frac{(1+C_V)}{4\pi} V^{-1} i \partial_+ V + j_+ \approx 0, \quad (2.47)$$

where

$$j_+ = \psi_2^{(0)} \psi_2^{(0)\dagger} + \{b_+^{(0)}, c_+^{(0)}\}, \quad \partial_- j_+ = 0. \quad (2.48)$$

One readily checks that this constraint is first class (vanishing central charge).

In terms of Ω_+ , the BRST current (2.43) is seen to take the standard form expected from general considerations:¹¹

$$J_+^{(B)} = \text{tr} \left[c_+^{(0)} \Omega_+ - \frac{1}{2} b_+^{(0)} \{c_+^{(0)}, c_+^{(0)}\} \right], \quad \partial_- J_+^{(B)} = 0, \quad (2.49)$$

and hence the associated charge is nilpotent.

3. Nonlocal Decoupled Formulation of QCD_2

The partition function (2.23) is particularly useful in the strong coupling regime. Following the ideas of Ref. 8, we now obtain an alternative, nonlocal representation useful in the weak coupling regime. To this end we make use of the identity

$$\begin{aligned} & \exp \left[\frac{i}{4e^2} \int \text{tr} \frac{1}{2} [\partial_+ (Vi\partial_- V^{-1})]^2 \right] \\ &= \int \mathcal{D}E \exp \left[-i \int \text{tr} \left[\frac{1}{2} E^2 + \frac{E}{2e} \partial_+ (Vi\partial_- V^{-1}) \right] \right] \end{aligned} \quad (3.1)$$

and make the change of variable

$$\partial_+ E = e \left(\frac{1 + C_V}{2\pi} \right) \beta^{-1} i \partial_+ \beta \quad (3.2)$$

or

$$E = e \left(\frac{1 + C_V}{2\pi} \right) (\partial_+)^{-1} \beta^{-1} i \partial_+ \beta. \quad (3.3)$$

The Jacobian associated with this change of variables is

$$\mathcal{D}E = \exp(-iC_V \Gamma[\beta]) \mathcal{D}\beta. \quad (3.4)$$

Making use of the above results, the partition function (2.23) reads

$$\begin{aligned} Z &= Z_F^{(0)} Z_{\text{gh}}^{(0)} \int \mathcal{D}V \int \mathcal{D}\beta \\ &\times \exp \left\{ -i(1 + C_V) \left[\Gamma[V] + \Gamma[\beta] - \frac{1}{4\pi} \int \text{tr}(\beta^{-1} \partial_+ \beta V \partial_- V^{-1}) \right] \right\} \\ &\times \exp(i\Gamma[\beta]) \exp \left\{ i \left(\frac{1 + C_V}{2\pi} \right)^2 e^2 \int \frac{1}{2} \text{tr}[\partial_+^{-1}(\beta^{-1} \partial_+ \beta)]^2 \right\}. \end{aligned} \quad (3.5)$$

Now, using the Polyakov–Wiegmann identity (2.38) and making the change of variable $V \rightarrow \beta V = \tilde{V}$, we are left with

$$Z = Z_F^{(0)} Z_{\text{gh}}^{(0)} Z_{\tilde{V}} Z_{\beta}, \quad (3.6)$$

where

$$Z_{\beta} = \int \mathcal{D}\beta \exp \left\{ i\Gamma[\beta] + i \left(\frac{1 + C_V}{2\pi} \right)^2 e^2 \int \frac{1}{2} \text{tr}[\partial_+^{-1}(\beta^{-1} \partial_+ \beta)]^2 \right\}, \quad (3.7)$$

$$Z_{\tilde{V}} = \int \mathcal{D}\tilde{V} \exp[-i(1 + C_V)\Gamma[\tilde{V}]]. \quad (3.8)$$

Expression (3.6) agrees with that of Ref. 8.

BRST invariance of the $(F - \text{gh} - \tilde{V})$ sector

The product $Z_F^{(0)} Z_{\text{gh}}^{(0)} Z_{\tilde{V}}$ is invariant under the BRST transformations (2.27) and (2.41) with the substitution $V \rightarrow \tilde{V}$, and $\Delta_{\mp}(\tilde{V})$ now given by

$$\Delta_{-}(\tilde{V}) = - \left(\frac{1 + C_V}{4\pi} \right) \tilde{V} i \partial_{-} \tilde{V}^{-1} \quad (3.9)$$

and

$$\Delta_{+}(\tilde{V}) = - \left(\frac{1 + C_V}{4\pi} \right) \tilde{V}^{-1} i \partial_{+} \tilde{V}, \quad (3.10)$$

implying the conservation of the corresponding Noether currents

$$\tilde{J}_{-}^{(B)} = \text{tr } c_{-}^{(0)} \left[\psi_1^{(0)} \psi_1^{(0)\dagger} + \frac{1}{2} \{b_{-}^{(0)}, c_{-}^{(0)}\} - \left(\frac{1 + C_V}{4\pi} \right) \tilde{V} \partial_{-} \tilde{V}^{-1} \right], \quad (3.11)$$

$$\tilde{J}_{+}^{(B)} = \text{tr } c_{+}^{(0)} \left[\psi_2^{(0)} \psi_2^{(0)\dagger} + \frac{1}{2} \{b_{+}^{(0)}, c_{+}^{(0)}\} - \left(\frac{1 + C_V}{4\pi} \right) \tilde{V}^{-1} \partial_{+} \tilde{V} \right]. \quad (3.12)$$

One easily verifies that the corresponding charges are nilpotent. Indeed, following again the procedure of Ref. 15, one gauges the left- and right-handed sector as in the conformal sector ($e \rightarrow \infty$) of the local formulation, and shows (with V replaced by $\tilde{V}$) that the partition function of the $F - \text{gh} - \tilde{V}$ sector remains invariant under this gauging. This implies the existence of the two first class constraints

$$\begin{aligned} \tilde{\Omega}_{-} &\equiv \psi_1^{(0)} \psi_1^{(0)\dagger} + \{b_{-}^{(0)}, c_{-}^{(0)}\} - \frac{1 + C_V}{4\pi} \tilde{V} i \partial_{-} \tilde{V}^{-1} \approx 0, \\ \tilde{\Omega}_{+} &\equiv \psi_2^{(0)} \psi_2^{(0)\dagger} + \{b_{+}^{(0)}, c_{+}^{(0)}\} - \frac{1 + C_V}{4\pi} \tilde{V}^{-1} i \partial_{+} \tilde{V} \approx 0. \end{aligned} \quad (3.13)$$

In terms of these constraints, the currents (3.11) and (3.12) take the standard form of BRST currents associated with a first class constraint algebra:¹¹

$$\tilde{J}_{\pm}^{(B)} = \text{tr} \left[c_{\pm} \tilde{\Omega}_{\pm}^{(0)} - \frac{1}{2} b_{\pm}^{(0)} \{c_{\pm}^{(0)}, c_{\pm}^{(0)}\} \right]. \quad (3.14)$$

It is important to note that these BRST currents of the nonlocal formulation correspond to the currents (2.40) and (2.49) of the local one.

To conclude this section let us remark that there exist further BRST-like symmetries, which, however, are not generated by nilpotent charges. To take an example, consider the $\text{gh} - \tilde{V} - \beta$ sector. The partition function $Z_{\text{gh}} Z_{\tilde{V}} Z_{\beta}$ is readily seen to be invariant under the symmetry transformation

$$\begin{aligned} \tilde{V} \delta \tilde{V}^{-1} &= -\epsilon c_{-}^{(0)}, \\ \beta \delta \beta^{-1} &= -\epsilon c_{-}^{(0)}, \\ \delta c_{-}^{(0)} &= \frac{\epsilon}{2} \{c_{-}^{(0)}, c_{-}^{(0)}\}, \quad \delta c_{+}^{(0)} = 0, \\ \delta b_{-}^{(0)} &= -\epsilon \frac{1 + C_V}{4\pi} \tilde{V} i \partial_{-} \tilde{V}^{-1} + \epsilon \{b_{-}^{(0)}, c_{-}^{(0)}\} + \epsilon \Delta_{-}[\beta], \end{aligned} \quad (3.15)$$

where $\Delta_-[\beta]$ is given by

$$\Delta_- (\beta) = \left(\frac{1+C_V}{2\pi} \right)^2 e^2 [\beta \partial_+^{-2} (\beta^{-1} i \partial_+ \beta) \beta^{-1}] + \frac{1}{4\pi} \beta i \partial_- \beta^{-1}. \quad (3.16)$$

The corresponding Noether current is found to be

$$\tilde{J}_- = \text{tr} \left[c_-^{(0)} \Omega - \frac{1}{2} b_-^{(0)} \{ c_-^{(0)}, c_-^{(0)} \} \right], \quad (3.17)$$

where

$$\begin{aligned} \Omega \equiv & - \left(\frac{1+C_V}{2\pi} \right)^2 e^2 \beta (\partial_+^{-2} (\beta^{-1} i \partial_+ \beta)) \beta^{-1} + \frac{1}{4\pi} \beta i \partial_- \beta^{-1} \\ & - \left(\frac{1+C_V}{4\pi} \right) \tilde{V} i \partial_- \tilde{V}^{-1} + \{ b_-^{(0)}, c_-^{(0)} \}. \end{aligned} \quad (3.18)$$

By gauging the right-handed ghost, and $\tilde{V}$ sector as in (2.37), (with $V \rightarrow \tilde{V}$), supplemented by the transformations

$$\begin{aligned} \Gamma[\beta] & \rightarrow \Gamma[\beta] - \frac{i}{4\pi} \int \text{tr} W_+ \beta \partial_- \beta^{-1} = \Gamma[\omega\beta] - \Gamma[\beta], \\ \beta^{-1} \partial_+ \beta & \rightarrow \beta^{-1} (\partial_+ - i W_+) \beta = (\omega\beta)^{-1} \partial_+ (\omega\beta) \end{aligned} \quad (3.19)$$

in the β sector, one finds that expression (3.18) is constrained to vanish:

$$\Omega \approx 0. \quad (3.20)$$

As pointed out in Ref. 8, this constraint is, however, not first class with respect to the constraints (3.13), and as a consequence, the BRST charges associated with the currents (3.14) already represent a complete set.

The role of the constraint (3.20) is best appreciated by rewriting the partition function Z_β in (3.7) with the aid of an auxiliary field C_- as⁸

$$Z_\beta = \int \mathcal{D}C_- \mathcal{D}\beta e^{iS[\beta, C_-]}, \quad (3.21)$$

where

$$S[\beta, C_-] = \Gamma[\beta] + \int \text{tr} \left[\frac{1}{2} (\partial_+ C_-)^2 + \left(\frac{1+C_V}{2\pi} \right) e(C_- \beta^{-1} i \partial_+ \beta) \right]. \quad (3.22)$$

Gauging $Z_{gh} Z_{\tilde{V}} Z_\beta$ as described before ($\partial_+ C_-$ and C_- remain unchanged), one arrives at the constraint

$$\left(\frac{1+C_V}{2\pi} \right) e \beta i C_- \beta^{-1} + \left(\frac{1+C_V}{4\pi} \right) \tilde{V} i \partial_- \tilde{V}^{-1} - \frac{1}{4\pi} \beta i \partial_- \beta^{-1} + \{ b_-^{(0)}, c_-^{(0)} \} = 0. \quad (3.23)$$

This constraint determines C_- as a function of the other fields. Using the equation of motion for C_- ,

$$\partial_+^2 C_- + \left(\frac{1 + C_V}{2\pi} \right) e\beta^{-1} \partial_+ \beta = 0,$$

one then formally arrives at constraint (3.20).

4. Discussion

The main objective of this paper was to further elucidate the interesting analysis of Ref. 8 by supplementing it with a study of the BRST symmetries as well as the construction of the corresponding BRST currents. At the same time we have given a concise derivation of those results of Ref. 8 which provide the basis for this analysis. In particular, all constraints, including also Gauß's law, were obtained in a purely path integral framework, following the method of Ref. 15. Furthermore, the so-called "local" formulation of Ref. 8 was obtained directly, without the introduction of further degrees of freedom subject to second class constraints. This served to "streamline" somewhat the manipulations in Ref. 8.

We have seen that the conformal invariance of the pure, gauged fermionic partition function, described by a WZW type theory,³ is broken by the presence of the Yang-Mills action, which implies the presence of a coupling constant carrying dimensions. We have nevertheless seen the BRST currents to be either left- or right-moving as in a conformally invariant model. This is remarkable and is at the heart of the claim in Ref. 8 on the exact integrability of QCD₂. Similar observations were made in this reference from a somewhat different point of view.

The study of BRST currents is important if one wants to obtain a complete characterization of the physical Hilbert space. It turns out that the BRST charges associated with the currents (3.14) of the "nonlocal" formulation provide here the most appropriate starting point. The implications of our BRST analysis for the structure of the Hilbert space are discussed in a recent paper by E. Abdalla and one of the authors of this paper.¹⁶

Acknowledgments

One of the authors (K. D. R.) would like to thank the Physics Department of the Universidad Nacional de La Plata for the kind hospitality extended to him, as well as the DAAD for the financial support making this stay possible. He would also like to thank Elcio Abdalla for very useful discussions.

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